Assignment NO:-6

Q.1]

```
void fun(int n){
    if(n==0){
        return;
    }
    s.o.p(n+" ");
    f(n-1);
    s.o.p((n+1)+" ")
```

call = f(3);

f(0) X
f(1) X
f(2) X
f(3) X

callstack

op

3 321234

2

1

```
void fun(3){
    if(n==0){
        return;
    }
    s.o.p(n+" ");  → 3
    f(2)
    s.o.p(n+1)+" ") → 3+1=4
```

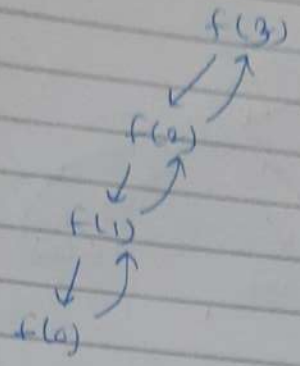
```
void fun(2){
    if(n==0){
        return;
    }
    s.o.p(n+" ");  → 2
    f(1)
    s.o.p(n+1) → 2+1=3
```

```
void fun(1){
    if(n==0){
        return;
    }
    s.o.p(n+" ");  → 1
    f(0)
    s.o.p(1+1)+" ");  → 2
```

```
void fun(0){
    if(0==0){
        return;
    }
```

Page No:
Date:

Recursive tree



op:
Total calls = 4
op = 3-2-1-2-3-4

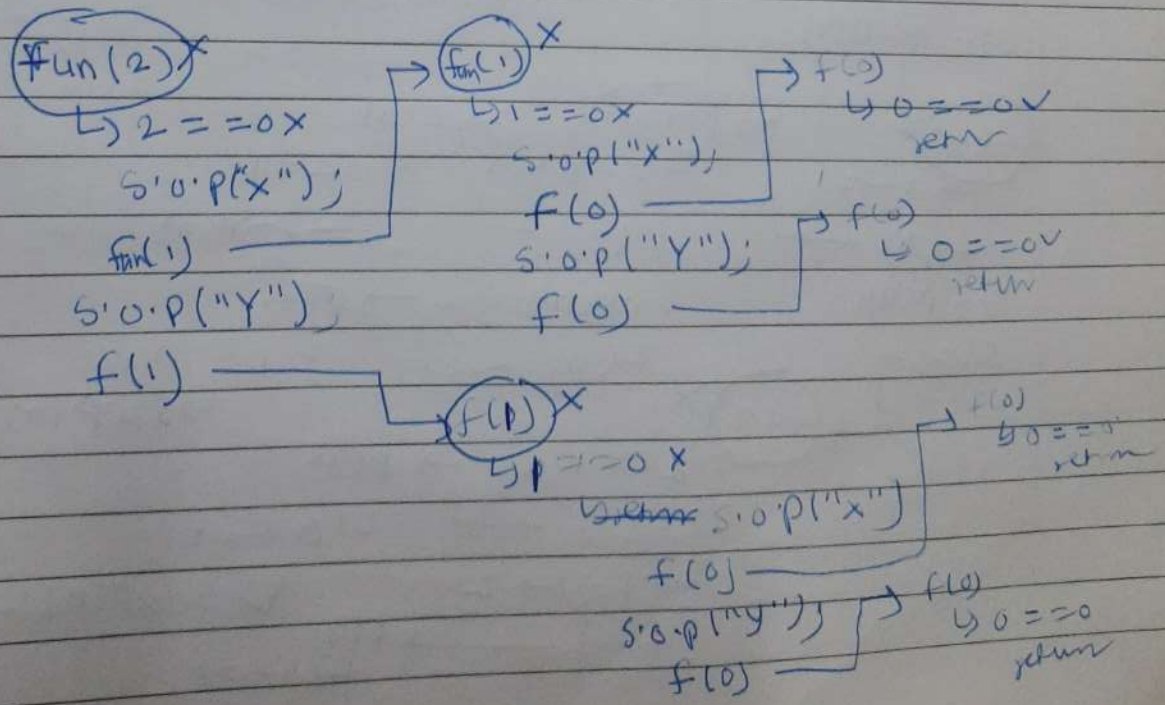
Q.2]
```
void fun(int n){
    if(n==0){
        return;
    }
    s.o.p("x");
    f(n-1);
    s.o.p("Y");
    f(n-1);
}
```

f(0) X
f(0) X
f(1) X
f(0) X
f(0) X
f(1) X
f(2) X

Call stack

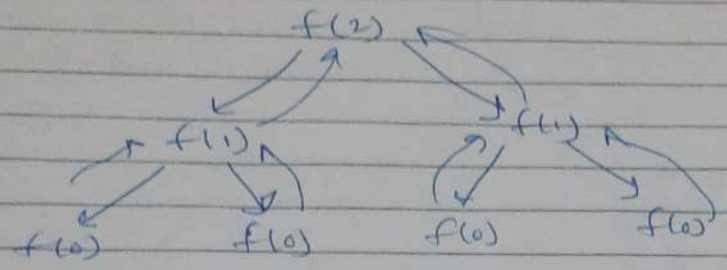
op
X X Y Y X Y
Total calls = 7.

call: fun(2)



Page No:
Date:

Recursive Tree



Q.3 Predict the output

```
void fun(int n){
    if(n<=1){
        s.o.p(n+" ");
        return;
    }
    f(n-2)
    s.o.p(n+" ");
    f(n-2);
}
```

call: fun(4)

	op
	0
	2
f(0)✗	0
f(0)✗	4
f(2)✗	0
f(0)✗	
f(0)✗	
f(2)✗	
f(4)✗	

f(4)✗
↳ 4<=1
f(2)
s.o.p(n)
f(2)

f(2)✗
↳ 2<=1
f(0)
s.o.p(n); → 2
f(0)

f(0)
↳ 0<=1
↳ s.o.p(n+" ");

f(0)
↳ 0<=1
↳ s.o.p(n); → 0

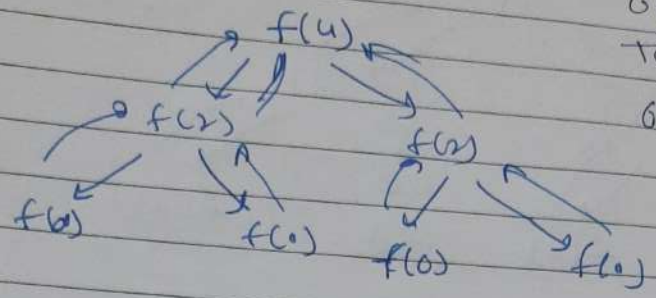
f(2)✗
↳ 2<=1 ✗
f(0)
s.o.p(n) → 2
f(0)

f(0)
↳ 0<=1
↳ s.o.p(n); → 0

f(0)
↳ 0<=1
↳ s.o.p(n) → 0

Page No:
Date.

Recursive Tree



OP =
Total calls = 7
OP = 0-2-0-4-0.

Q.4

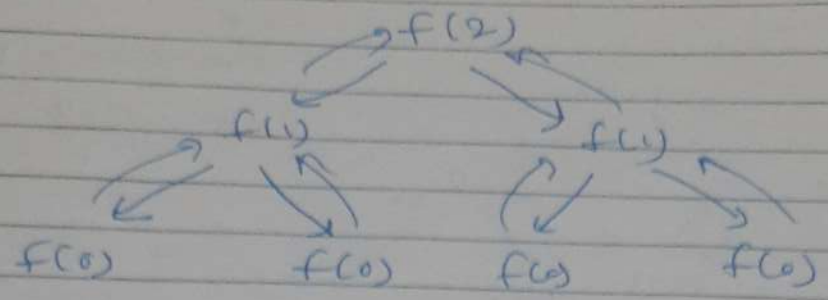
```
void f(int n){
    if(n==0){
        return;
    }
    f(n-1);
    s.o.p(n*n+" ");
    f(n-1);
}
```

f(0)✗	OP
f(0)✗	1
~~f(1)✗~~	4
f(1)✗	1
f(0)✗	
f(0)✗	OP = 1-4-1
f(1)✗	calls = 7.
f(2)✗	

call stack.

call:
f(2)

f(2)✗
↳ 2==0✗
f(1) → f(1)
↳ 1==0✗
f(0) → f(0)
↳ 0==0
return.
s.o.p(1*1+" ") → 1
f(0) → f(0)
↳ 0==0
return.
s.o.p(2*2+" ");→ 4
f(1) → f(1)✗
↳ 1==0✗
f(0) → f(0)
↳ 0==0
s.o.p(n*n) → 1
f(0) → f(0)
↳ 0==0✓



Q.5 Predict the output

```
void fun (int n){
    if (n<=0)
        return;
    s.o.p(n + " ");
    fun(n-2);
    s.o.p(n + " ");
}
```

call = f(5);

	op
	5
	3
	1
	-1
f(-1)×	3
f(1)×	5
f(3)×	op 5 3 1 -1 3 5
f(5)×	

callstack

f(5) ×
↳ 5<=0 ×
s.o.p(n) → 5
f(3)
s.o.p(n) → 5

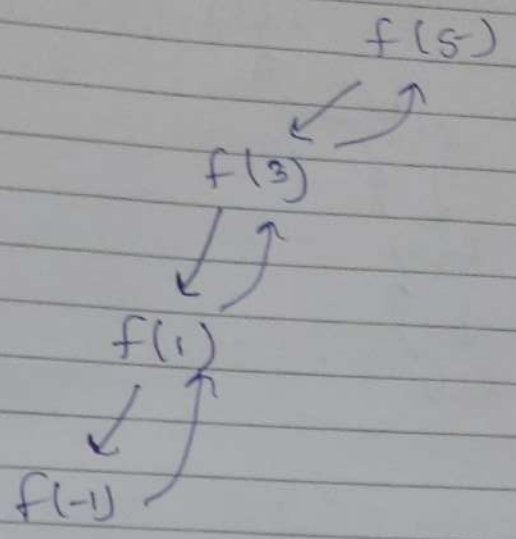
f(3) ×
↳ 3<=0 ×
s.o.p(n) → 3
f(1)
s.o.p(n) → 3

f(1) ×
↳ 1<=0 ×
s.o.p(n) → 1
f(-1) -
s.o.p(n) → -

f(-1)
↳ -1<=0

op =
Total calls = 4
op = 5-3-1-1-3-5

~~f(-1)~~
~~↳ -1 <= 0~~



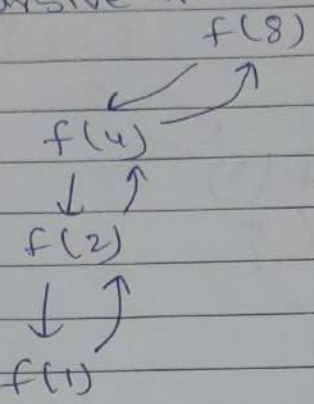
Q5 Predict the output

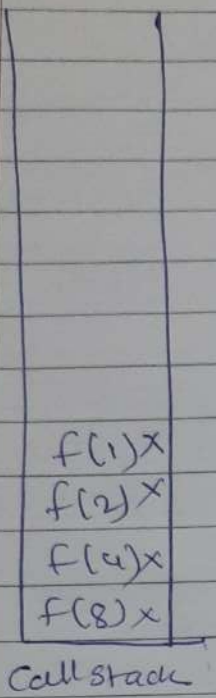
```
Void fun(int n){
    if (n <= 1){
        s.o.p(n+" ");
        return;
    }
    s.o.p(n+" ")
    fun(n/2);
    s.o.p(n+" ");
}

call = fun(8);
```

Page No:
Date.

OP = 8-4-2-1-2-4-8.

Recursive tree





f(8) ×
↳ 8 <= 1 ×
S.O.P(8);
f(4)
S.O.P(8) → 8

f(4) ×
↳ 4 <= 1 ×
S.O.P(4); → 4
f(2)
S.O.P(4) → 4

f(2) ×
↳ 2 <= 1 ×
S.O.P(2); → 2
f(1)
S.O.P(2) → 2

f(1)
↳ 1 <= 1 ✓
S.O.P(1) → 1

OP =
Total calls = 4
OP = 8-4-2-1-2-4-8

Q.7 Predict the output.

Call = f(3);

```
Void fun(int n) {
    if (n<=0)
        return;

    S.O.P("A" + n + " ");
    f(n-1);
    S.O.P("B" + (n*2) + " ");
}
```

f(0)✗
f(1)✗
f(2)✗
f(3)✗

Callstack

OP: A3_A2_A1_B1_B4_B9

f(3)
↓↑
f(2)
↓↑
f(1)
↓↑
f(0)

f(3)✗
↳ 3<=0 ✗
S.O.P("A"+3+" ");
f(2)
S.O.P("B"+9+" ");

f(2)✗
↳ 2<=0 ✗
S.O.P("A"+2+" ");
f(1)
S.O.P("B"+4+" ");

f(1)✗
↳ 1<=0 ✗
S.O.P("A"+1+" ");
f(0)
S.O.P("B"+2+" ");

f(0)
↳ 0<=0 ✓

OP =
Total calls = 4
OP = A3_A2_A1_B1_B4_B9

Q.8 Predict the output

call = f(4)

```
void fun(int n) {
    if (n<=0)
        return;

    f(n-1);
    s.o.p(n+" ");
    f(n-2);
}
```

f(0) ✗	f(0) ✗
f(-1) ✗	f(-1) ✗
f(0) ✗	f(0) ✗
f(1) ✗	f(1) ✗
f(2) ✗	f(2) ✗
f(3) ✗	f(0) ✗
f(4) ✗	f(1) ✗

call stack

? op = 1 2 3 4 1 2

f(4) ✗
↳ 4<=0 ✗
f(3)
s.o.p(n+" ") → 4
f(2)

f(3) ✗
↳ 3<=0 ✗
f(2)
s.o.p(n+" ") → 3
f(1)

f(2) ✗
↳ 2<=0 ✗
f(1)
s.o.p(2) → 2
f(0)
f(0)
↳ 0<=0 ✓

f(1) ✗
↳ 1<=0 ✗
f(0)
s.o.p(1) → 1
f(-1)

f(-1)
↳ -1<=0 ✓

f(0)
↳ 0<=0 ✓

f(1)
↳ 1<=0 ✗
f(0)

f(0)
↳ 0<=0 ✓

f(2) ✗
↳ 2<=0 ✗
f(1)
s.o.p(2) → 2
f(0)

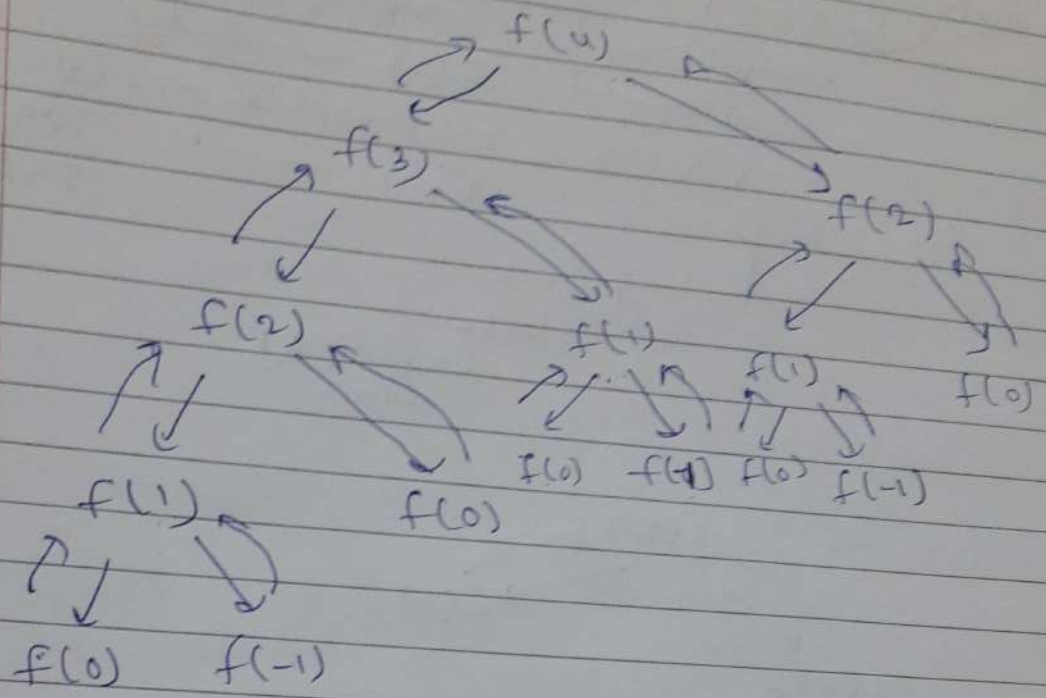
f(1) ✗
↳ 1<=0 ✗
f(0)
s.o.p(1) → 1
f(-1)

f(0)
↳ 0<=0

f(-1)
↳ -1<=0 ✗

f(0)
↳ 0<=0 ✗

Recursive Tree:-



Total execution calls = 13
OP = 1_2_3_4_1_2.

Q.9] Predict the output

```
Void fun(int n){
    if (n<=1){
        S.o.p(n+" ");
        return;
    }
    fun(n-1);
    S.o.p(n+" ");
    f(n-2);
}
```

Calls: f(3)

	~~OP = 3_1~~
	OP = 1_2_0_3
	1
f(1) X	
f(0) X	
f(1) X	
f(2) X	
f(3) X	

~~fun~~ f(3)

↳ 3 <= 1 ×

~~S.O.P(3) → 3~~

f(2)

S.O.P(3) → 3

f(1)

↳ f(1)

↳ 1 <= 1

S.O.P(n) → 1

return

f(2)

↳ 2 <= 1 ×

S.O.P(2) → 2

f(1)

S.O.P(2) → 2

f(0)

f(1)

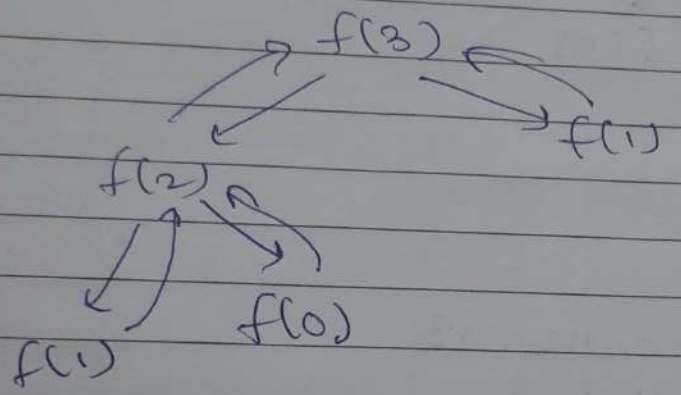
↳ 1 <= 1

S.O.P(n) → 1

f(0)

↳ 0 <= 1 ✓

S.O.P(n) → 0



~~Cat~~ O.P:

Total calls: 5

OP = 1-2-0-3-1

Q.10 Count Recursive Calls.

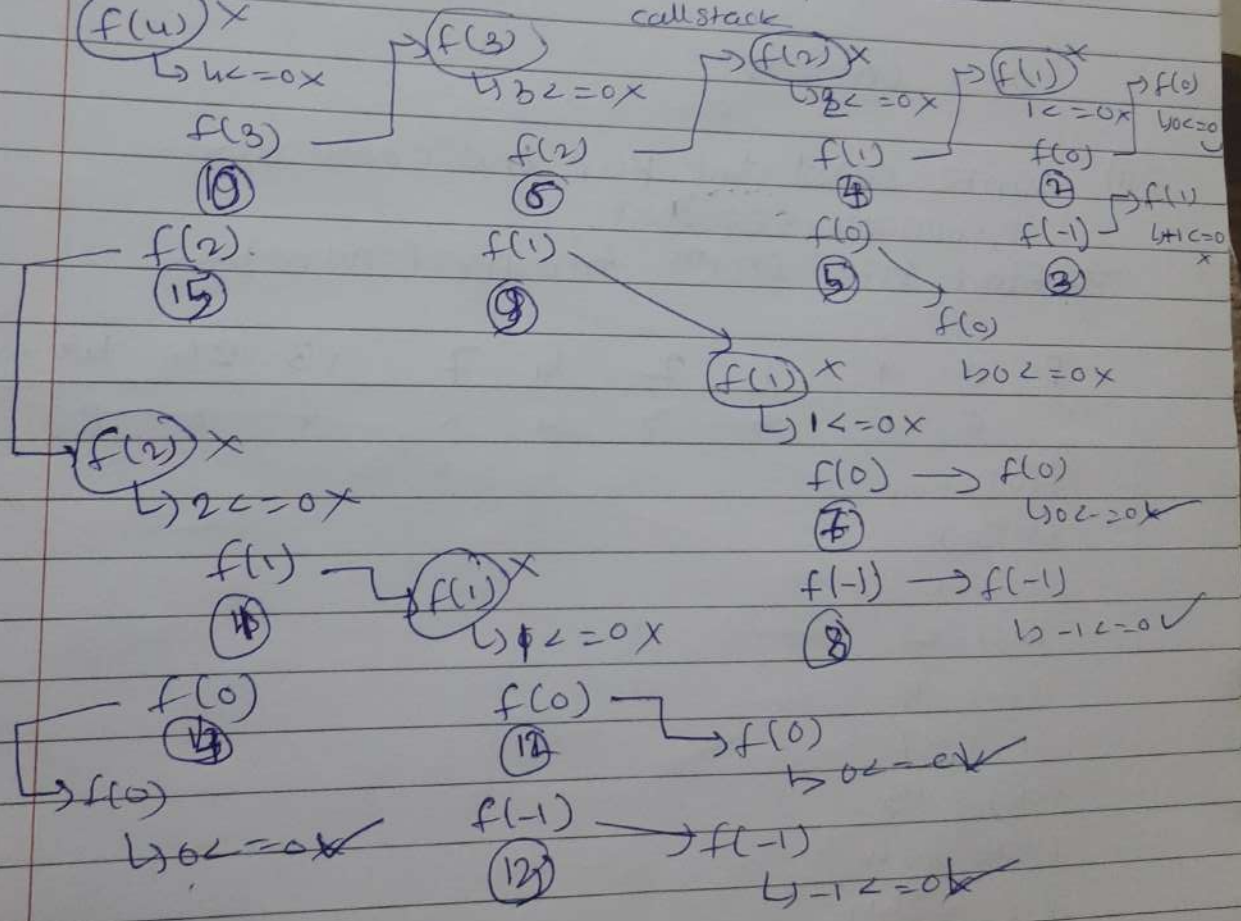
```
int cnt=1;
void fun(int n){
    if (n<=0)
        return;

    fun(n-1);
        cnt++;
    fun(n-2);
        cnt++;
}
```

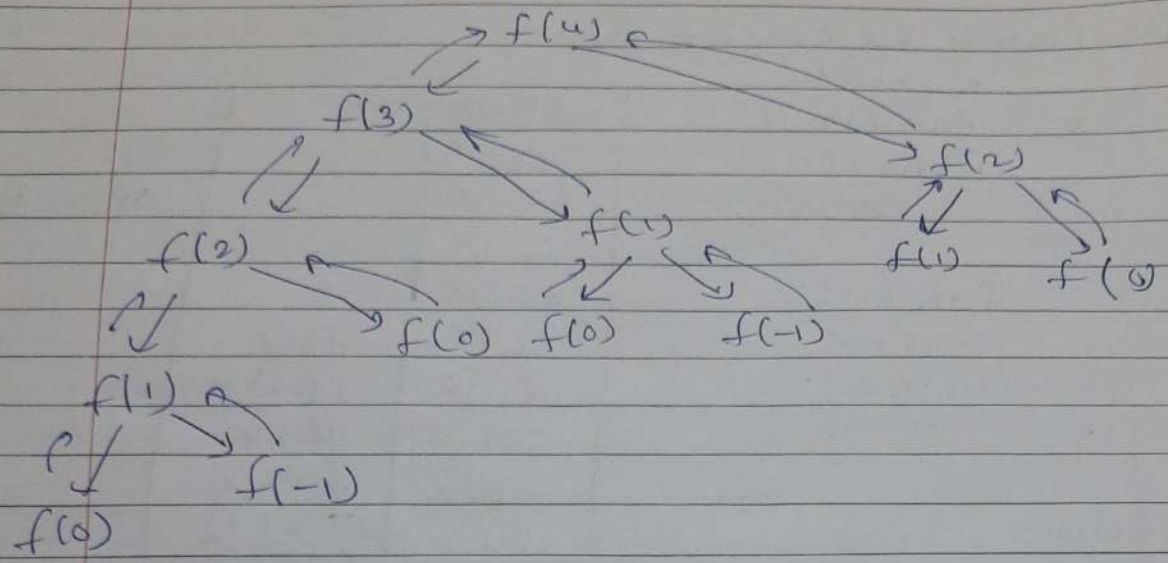
calls: fun(4)

f(-1) ✗	
f(0) ✗	
f(1) ✗	
f(0) ✗	
f(-1) ✗	
~~f(0)~~ ✗	f(0)
~~f(1)~~ ✗	f(-1) ✗
~~f(2)~~ ✗	f(0) ✗
~~f(3)~~ ✗	f(1) ✗
f(4) ✗	f(2) ✗

callstack

op = 15





op = cnt = 15

Q.11 write and dry Run the code for Tribonacci sequence.

→ find the $f(6)^{th}$ tribonacci number

0	1	1	2	4	7	13	24	44 ...
0	1	2	3	4	5	6	7	8

$f(0) = 0$
$f(1) = 1$
$f(2) = 1$
$f(3) = 2$
$f(4) = 4$
$f(5) = 7$
$f(6) = 13$
$f(7) = 24$
$f(8) = 44$

```
psvm(){
    int n = 6;
    int cnt = f(n);
        S.O.P(cnt);
}

int fun(int n){
    if(n<=0){
        return 0;
    }
    if(n==1){
        return 1;
    }
    int last = f(n-1);
    int s_last = f(n-2);
    int t_last = f(n-3);
    return last+s_last+t_last;
}
```

Page No:
Date:

f(6) $\boxed{13}$
↳ 6<=0 ×
6==1 ×
f(5) $\boxed{7}$
f(4) $\boxed{4}$
f(3) $\boxed{2}$
7+4+2=13

f(5) $\boxed{7}$
↳ 5<=0 ×
5==1 ×
f(4) $\boxed{4}$
f(3) $\boxed{2}$
f(2) $\boxed{1}$
4+2+1=7

f(4) $\boxed{4}$
↳ 4<=0 ×
4==1 ×
f(3) $\boxed{2}$
f(2) $\boxed{1}$
f(1) $\boxed{1}$
2+1+1=4

f(3) × $\boxed{2}$
↳ 3<=0 ×
3==1 ×
f(2) $\boxed{1}$
f(1) $\boxed{1}$
f(0) $\boxed{0}$
1+1+0=$\boxed{2}$
f(0)
↳ 0<=0
↳ 0

f(2) × $\boxed{1}$
↳ 2<=0 ×
2==1 ×
f(1) $\boxed{1}$
f(0) $\boxed{0}$ → f(0)
f(-1) $\boxed{0}$ ↳ 0<=0
1+0+0=$\boxed{1}$ ↳ 0
f(-1)
↳ -1<=0
↳ 0

f(1) f
↳ 1<=0 ×
1==1
↳ 1
~~f(0)~~

f(0)
↳

f(1)
↳ 1==1
↳ 1

f(2) $\boxed{1}$
↳ 2<=0 ×
2==1 ×
f(1) $\boxed{1}$
f(0) $\boxed{0}$ → f(0)
f(-1) $\boxed{0}$ ↳ 0<=0 ✓
1+0+0=$\boxed{1}$ ↳ $\boxed{0}$
f(-1)
↳ -1<=0
↳ 0

f(1)
↳ 1<=0 ×
1==1
↳ $\boxed{1}$

f(1)
↳ 1<=0 ×
1==1
↳ $\boxed{1}$

Page No:
Date.

$f(3)$ $\boxed{2}$

↳ $3 <= 0$ ✗

$3 == 1$ ✗

$f(2)$ $\boxed{1}$

$f(1)$ $\boxed{1}$

$f(0)$ $\boxed{0}$

$1+1+0 = 2$

$f(2)$ $\boxed{1}$

↳ $2 <= 0$ ✗

$2 == 1$ ✗

$f(1)$ $\boxed{1}$

$f(0)$ $\boxed{0}$ → $f(0)$

↳ $0 <= 0$ ✓

$f(-1)$ $\boxed{0}$ → $f(-1)$ → $-1 <= 0$ ✓

$1+0+0 = 1$

$f(1)$

↳ $1 <= 0$ ✗

$1 == 1$

$f(0)$

↳ $0 <= 0$ ✓

$f(1)$

↳ $1 <= 0$ ✗

$1 == 1$ ✓

$f(2)$ $\boxed{1}$

↳ $2 <= 0$ ✗

$2 == 1$

$f(1)$ $\boxed{1}$ → $f(1)$

↳ $1 <= 0$ ✗

$1 == 1$

$f(0)$ $\boxed{0}$

$f(-1)$ $\boxed{0}$

$1+0+0 = 1$

$f(0)$

↳ $0 <= 0$

$f(-1)$

↳ $-1 <= 0$ ✓